Liam Hopkins

DS 5030

Lab 6

Market Testing Write Up

The goal of this analysis was to find the price that maximizes revenue per customer with only a small number of market tests. To do so, I compared the estimated revenue per customer for different prices from $5 to $100. I started by doing a coarse scan, testing a handful of evenly spaced prices and using their corresponding 95% confidence interval of the estimated revenue per customer to plot a picture of the distribution. Based on the information I found, I began doing hypothesis tests on the difference between the revenue per customer of prices that appeared closest to the optimal value in the plot. These hypothesis tests were done by calculating the 95% confidence interval of these differences and checking if they contained 0. If I found the difference between two prices was statistically significant, I would test values closer to the price which was found to have a larger revenue. If my test failed to reject the null hypothesis, I would take another market test for one of the prices, pool the samples for that price, and conduct the test again. Once I found myself continuing to pool without the CI bounds noticeably shrinking closer to 0, or I felt confident in which price was largest, I would stop.

For my coarse scan, I did 7 market tests on price values starting at $5 and going to $100 with a step size of 15. Figure 1 shows the calculated confidence intervals for the revenue per customer of each of the prices.

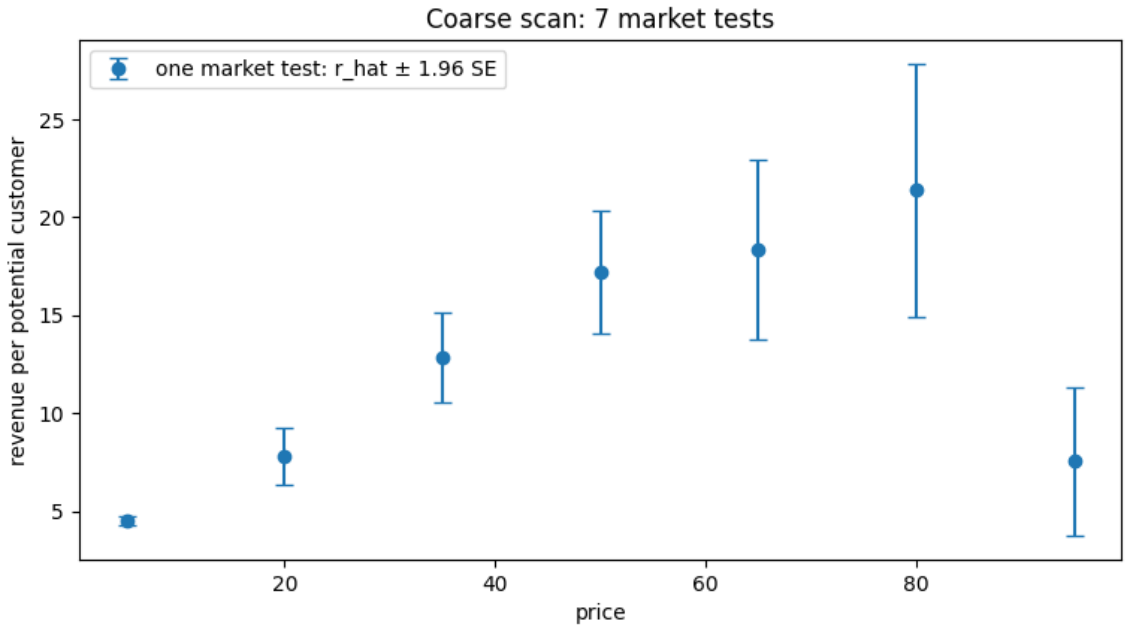


Figure 1: 95% Confidence intervals of the 7 Tested Prices

Based on the confidence intervals in the plot, the range of prices from $65 to $80 appears to be closest to the optimal price. Prices less than $65 appear to drop in revenue per customer as they shrink toward 0, and there appears to be a large drop in revenue for prices larger than $80. Because of this, I chose $65 and $80 as my first two prices to test. As I continued to pool more data, I noticed the upper bound of the confidence intervals for the difference decreasing toward 0. This was my sign to keep market testing. After a few tests, 0 fell out of the interval and both bounds were negative, so I was able to conclude that $80 had a higher revenue than $65. I then began testing again with $75 instead of $65. The results of this test were the same, as I found $80 had a higher revenue than $75 after a few tests.

After these results, $80 is my final recommended price. In total, I ran 14 market tests across my coarse scan and my testing. At this price, I estimate that the 95% confidence interval for the revenue per customer is (20.92, 24.57). To make this choice, I used my confidence intervals to tell me when to stop. The confidence intervals for the difference between the revenue per customer of $80 and smaller prices near it showed $80 had a larger revenue, as 0 fell out of the intervals after pooling several market tests and both bounds were negative. Also, my intervals in the plot from my coarse scan showed a large drop in revenue for values larger than $80, meaning it would not be worth the cost to test those values.